Project2

March 12, 2026

1 Project 2: Predicting Restaurant Inspection Score Change

Question: Given a restaurant's most recent routine inspection, construct a valid prediction interval for how much its inspection score will change at its next routine inspection.

Methodology: - **Part 1 (Selective Inference):** Knockoffs for FDR-controlled feature selection + FCR confidence intervals on selected features. - **Part 2 (Predictive Inference):** Three conformal prediction methods: - Rescaled residual conformal - Conformalized Quantile Regression (CQR) - Local kNN conformal - **Evaluation:** Coverage on 2019 holdout data

1.1 Setup

[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy import stats
import statsmodels.api as sm
from statsmodels.regression.quantile_regression import QuantReg
from sklearn.preprocessing import StandardScaler
import warnings
warnings.filterwarnings('ignore')

np.random.seed(42)

FCR_ALPHA = 0.1
COVERAGE_ALPHA = 0.05
FDR_LEVEL = 0.2
```

1.2 Step 0.1: Preprocessing

[2]:
```
raw_data = pd.read_csv('./restaurantdata.csv')
print(f'Raw data shape: {raw_data.shape}')
print(raw_data.dtypes)
```

```
Raw data shape: (53973, 17)
business_id                int64
business_name             object
business_address          object
business_city             object
```

```
business_state            object
business_postal_code      object
business_latitude        float64
business_longitude       float64
business_location         object
business_phone_number    float64
inspection_id             object
inspection_date           object
inspection_score         float64
inspection_type           object
violation_id              object
violation_description     object
risk_category             object
dtype: object
```

[3]:

```
df = raw_data.copy()

# (1) Non-missing inspection score
df = df[df['inspection_score'].notna()].copy()
df['inspection_score'] = pd.to_numeric(df['inspection_score'], errors='coerce')
df = df[df['inspection_score'].notna()].copy()

# (2) Parse date
df['inspection_date'] = pd.to_datetime(
    df['inspection_date'], format='%m/%d/%Y %I:%M:%S %p', errors='coerce'
)
df = df[df['inspection_date'].notna()].copy()
df['month'] = df['inspection_date'].dt.month
df['day']   = df['inspection_date'].dt.day
df['year']  = df['inspection_date'].dt.year

# (3) Replace empty risk_category with 'None'
df['risk_category'] = df['risk_category'].replace('', 'None')

df = df.rename(columns={
    'business_postal_code': 'zipcode',
    'business_latitude':    'latitude',
    'business_longitude':   'longitude'
})

df = df.sort_values(['business_id', 'inspection_date']).reset_index(drop=True)

data1 = df[['month', 'day', 'year', 'business_id', 'risk_category',
            'inspection_score', 'zipcode', 'latitude', 'longitude',
            'inspection_date']].copy()

print(f'data1 shape: {data1.shape}')
```

```
print(f'Unique inspections (business_id + date): {data1.
↪groupby(["business_id","inspection_date"]).ngroups}')
data1.head(10)
```

```
data1 shape: (39541, 10)
Unique inspections (business_id + date): 14031
```

[3]:

```
   month  day  year  business_id  risk_category  inspection_score zipcode  \
0     12   11  2017           19       Low Risk              94.0   94109
1     12   11  2017           19  Moderate Risk              94.0   94109
2      6    7  2018           19  Moderate Risk              96.0   94109
3     10    5  2016           24            NaN              98.0   94104
4     11    1  2017           24       Low Risk              98.0   94104
5      5    1  2019           24       Low Risk              98.0   94104
6      6   15  2018           31       Low Risk              96.0   94133
7      6   15  2018           31       Low Risk              96.0   94133
8      7    2  2019           31  Moderate Risk              94.0   94133
9      7    2  2019           31       Low Risk              94.0   94133

    latitude   longitude inspection_date
0  37.786848 -122.421547      2017-12-11
1  37.786848 -122.421547      2017-12-11
2  37.786848 -122.421547      2018-06-07
3  37.792888 -122.403135      2016-10-05
4  37.792888 -122.403135      2017-11-01
5  37.792888 -122.403135      2019-05-01
6  37.807155 -122.419004      2018-06-15
7  37.807155 -122.419004      2018-06-15
8  37.807155 -122.419004      2019-07-02
9  37.807155 -122.419004      2019-07-02
```

Each inspection can have multiple rows (one per violation). We aggregate risk categories into binary features.

[4]:

```
inspections = (
    data1
    .groupby(['business_id', 'inspection_date', 'month', 'day', 'year',
              'zipcode', 'latitude', 'longitude'])
    .agg(
        inspection_score = ('inspection_score', 'first'),
        has_low          = ('risk_category', lambda x: int('Low Risk' in x.
↪values)),
        has_moderate     = ('risk_category', lambda x: int('Moderate Risk' in x.
↪values)),
        has_high         = ('risk_category', lambda x: int('High Risk' in x.
↪values)),
        num_violations   = ('risk_category', 'count')
    )
```

```
    .reset_index()
    .sort_values(['business_id', 'inspection_date'])
    .reset_index(drop=True)
)

print(f'Unique inspections: {len(inspections)}')
print(f'Unique restaurants: {inspections["business_id"].nunique()}')
inspections.head(8)
```

```
Unique inspections: 7297
Unique restaurants: 2683
```

[4]:

```
   business_id inspection_date  month  day  year zipcode   latitude  \
0           19      2017-12-11     12   11  2017   94109  37.786848
1           19      2018-06-07      6    7  2018   94109  37.786848
2           24      2016-10-05     10    5  2016   94104  37.792888
3           24      2017-11-01     11    1  2017   94104  37.792888
4           24      2019-05-01      5    1  2019   94104  37.792888
5           31      2018-06-15      6   15  2018   94133  37.807155
6           31      2019-07-02      7    2  2019   94133  37.807155
7           45      2017-03-07      3    7  2017   94110  37.747114

    longitude  inspection_score  has_low  has_moderate  has_high  \
0 -122.421547              94.0        1             1         0
1 -122.421547              96.0        0             1         0
2 -122.403135              98.0        0             0         0
3 -122.403135              98.0        1             0         0
4 -122.403135              98.0        1             0         0
5 -122.419004              96.0        1             0         0
6 -122.419004              94.0        1             1         0
7 -122.413641              88.0        1             1         0

   num_violations
0               2
1               1
2               0
3               1
4               1
5               2
6               2
7               5
```

[5]:

```
# Find restaurants with more than one unique (lat, lon) pair
location_counts = (
    inspections
    .groupby('business_id')[['latitude', 'longitude']]
    .nunique()
)
```

```
# A restaurant "moved" if it has >1 unique lat OR >1 unique lon
moved = location_counts[(location_counts['latitude'] > 1) |␣
↪(location_counts['longitude'] > 1)]
print(f"Restaurants that appear to have moved: {len(moved)}")
print(moved)
```

```
Restaurants that appear to have moved: 0
Empty DataFrame
Columns: [latitude, longitude]
Index: []
```

For each restaurant with ≥ 2 inspections, create all consecutive pairs $(t_k \to t_{k+1})$. Target: $\Delta = \text{score}_{t_{k+1}} - \text{score}_{t_k}$. Restaurants with only one inspection are dropped.

[6]:
```
records = []

for business_id, grp in inspections.groupby('business_id'):
    grp = grp.sort_values('inspection_date').reset_index(drop=True)
    if len(grp) < 2:
        continue
    scores = grp['inspection_score'].values
    for k in range(len(grp) - 1):
        curr = grp.iloc[k]
        nxt  = grp.iloc[k + 1]

        # Historical mean score
        hist_score_mean = float(np.mean(scores[:k])) if k > 0 else scores[k]
        # Historical mean and std of consecutive deltas up to curr inspection
        diffs = np.diff(scores[:k+1]) if k > 0 else np.array([])
        hist_score_diff_mean = float(np.mean(diffs)) if len(diffs) > 0 else 0.0
        hist_delta_std       = float(np.std(diffs))  if len(diffs) > 1 else 0.0

        records.append({
            'business_id':        business_id,
            'pair_index':         k,
            'curr_date':          curr['inspection_date'],
            'next_date':          nxt['inspection_date'],
            'month':              curr['month'],
            'year':               curr['year'],
            'zipcode':            curr['zipcode'],
            'latitude':           curr['latitude'],
            'longitude':          curr['longitude'],
            'prev_score':         curr['inspection_score'],
            'has_low':            curr['has_low'],
            'has_moderate':       curr['has_moderate'],
            'has_high':           curr['has_high'],
            'num_violations':     curr['num_violations'],
```

```
            'time_gap':            (nxt['inspection_date'] -␣
↪curr['inspection_date']).days,
            'hist_score_mean':    hist_score_mean,
            'hist_score_diff_mean': hist_score_diff_mean,
            'hist_delta_std':      hist_delta_std,
            'delta':               nxt['inspection_score'] -␣
↪curr['inspection_score'],
        })

pairs = pd.DataFrame(records)
print(f'Total consecutive pairs: {len(pairs)}')
print(f'Restaurants contributing pairs: {pairs["business_id"].nunique()}')
pairs.head(6)
```

```
Total consecutive pairs: 4614
Restaurants contributing pairs: 2351
```

[6]:

	business_id	pair_index	curr_date	next_date	month	year	zipcode	\
0	19	0	2017-12-11	2018-06-07	12	2017	94109	
1	24	0	2016-10-05	2017-11-01	10	2016	94104	
2	24	1	2017-11-01	2019-05-01	11	2017	94104	
3	31	0	2018-06-15	2019-07-02	6	2018	94133	
4	45	0	2017-03-07	2017-09-14	3	2017	94110	
5	45	1	2017-09-14	2018-05-29	9	2017	94110	

	latitude	longitude	prev_score	has_low	has_moderate	has_high	\
0	37.786848	-122.421547	94.0	1	1	0	
1	37.792888	-122.403135	98.0	0	0	0	
2	37.792888	-122.403135	98.0	1	0	0	
3	37.807155	-122.419004	96.0	1	0	0	
4	37.747114	-122.413641	88.0	1	1	0	
5	37.747114	-122.413641	85.0	1	1	1	

	num_violations	time_gap	hist_score_mean	hist_score_diff_mean	\
0	2	178	94.0	0.0	
1	0	392	98.0	0.0	
2	1	546	98.0	0.0	
3	2	382	96.0	0.0	
4	5	191	88.0	0.0	
5	4	257	88.0	-3.0	

	hist_delta_std	delta
0	0.0	2.0
1	0.0	0.0
2	0.0	0.0
3	0.0	-2.0
4	0.0	-3.0

```
5              0.0    3.0
```

We split by `business_id` to ensure Set A and Set B are fully independent. Splitting by pair would allow the same restaurant to appear in both sets, violating independence for knockoffs and conformal prediction.

- **Set A** (50% of 2016-2018 restaurants): selective inference with knockoffs
- **Set B** (50% of 2016-2018 restaurants): predictive inference with conformal prediction
- **Test** (2019, first pair per restaurant): coverage evaluation

[7]:
```
# Separate 2019 test set first: use pairs where curr_date is in 2016-2018 and␣
↪next_date is in 2019
pairs_train = pairs[pairs['next_date'].dt.year <= 2018].copy()
pairs_test  = pairs[
    (pairs['curr_date'].dt.year <= 2018) & (pairs['next_date'].dt.year == 2019)
].copy()

print(f'Training pairs (2016-2018): {len(pairs_train)}')
print(f'Test pairs (last 2018 -> first 2019): {len(pairs_test)}')

# For test: keep only the FIRST 2019 pair per restaurant to avoid␣
↪within-restaurant correlation
pairs_test = (
    pairs_test
    .sort_values(['business_id', 'curr_date'])
    .groupby('business_id')
    .first()
    .reset_index()
)
print(f'Test pairs (first per restaurant): {len(pairs_test)}')

# Split training restaurants 50/50 into Set A and Set B
train_restaurants = pairs_train['business_id'].unique()
np.random.shuffle(train_restaurants)
n_A = int(len(train_restaurants) * 0.5)

rest_A = set(train_restaurants[:n_A])
rest_B = set(train_restaurants[n_A:])

set_A = pairs_train[pairs_train['business_id'].isin(rest_A)].copy()
set_B = pairs_train[pairs_train['business_id'].isin(rest_B)].copy()

print(f'\nSet A: {len(set_A)} pairs from {len(rest_A)} restaurants')
print(f'Set B: {len(set_B)} pairs from {len(rest_B)} restaurants')
print(f'Test:  {len(pairs_test)} pairs from {pairs_test["business_id"].
↪nunique()} restaurants')
```

```
Training pairs (2016-2018): 2519
Test pairs (last 2018 -> first 2019): 1819
```

```
Test pairs (first per restaurant): 1819

Set A: 1253 pairs from 897 restaurants
Set B: 1266 pairs from 898 restaurants
Test:  1819 pairs from 1819 restaurants
```

1.3 Step 0.2: Feature Engineering

Zipcode statistics are computed from only on Set A and applied as fixed constants to Set B and the test set. This prevents leakage from Set B's target values into the zipcode encoding.

[8]:
```
zipcode_stats = (
    set_A
    .groupby('zipcode')['prev_score']
    .agg(zipcode_avg_score='mean', zipcode_score_std='std')
    .reset_index()
)
global_std = set_A['prev_score'].std()
zipcode_stats['zipcode_score_std'] = zipcode_stats['zipcode_score_std'].
 ↪fillna(global_std)

# Bin zipcode_avg_score into 10-point intervals (50s→5, 60s→6, etc.)
zipcode_stats['zipcode_score_bin'] = (zipcode_stats['zipcode_avg_score'] // 10).
 ↪astype(int)

# Target encoding for zipcode_avg_delta
zipcode_delta_sum   = set_A.groupby('zipcode')['delta'].sum()
zipcode_delta_count = set_A.groupby('zipcode')['delta'].count()

def loo_zipcode_delta(row):
    z = row['zipcode']
    total = zipcode_delta_sum.get(z, 0)
    count = zipcode_delta_count.get(z, 0)
    if count <= 1:
        return set_A['delta'].mean()
    return (total - row['delta']) / (count - 1)

set_A = set_A.copy()
set_A['zipcode_avg_delta'] = set_A.apply(loo_zipcode_delta, axis=1)

# For Set B and test: use full Set-A zipcode mean
zipcode_delta_stats = (
    set_A.groupby('zipcode')['delta'].mean()
    .reset_index().rename(columns={'delta': 'zipcode_avg_delta'})
)
global_avg_delta = set_A['delta'].mean()
```

[9]:
```
FEATURE_COLS = [
    'prev_score',
    'has_low', 'has_moderate', 'has_high',
    'num_violations',
    'time_gap',
    'zipcode_score_bin',        # zipcode avg score floored to nearest 10
    'zipcode_avg_delta',
    'hist_score_diff_mean',
    'hist_delta_std',           # std of historical score deltas (volatility)
    'month_avg_delta',   # target encoding: mean delta across dataset for this␣
 ↪calendar month
    'year_avg_delta',    # target encoding: mean delta across dataset for this␣
 ↪year
]

# Compute all encodings from Set A
global_avg = set_A['prev_score'].mean()

set_A = set_A.merge(zipcode_stats, on='zipcode', how='left')
set_A['zipcode_avg_score'] = set_A['zipcode_avg_score'].fillna(global_avg)
set_A['zipcode_score_bin'] = (set_A['zipcode_avg_score'] // 10).astype(int)
set_A['zipcode_score_std'] = set_A['zipcode_score_std'].fillna(global_std)
set_A['score_deviation']   = set_A['prev_score'] - set_A['zipcode_avg_score']
set_A['hist_score_mean']      = set_A['hist_score_mean'].
 ↪fillna(set_A['prev_score'])
set_A['hist_score_diff_mean'] = set_A['hist_score_diff_mean'].
 ↪fillna(set_A['zipcode_avg_delta'])
set_A['hist_delta_std']       = set_A['hist_delta_std'].fillna(0.0)

# Temporal target encoding
month_delta_sum   = set_A.groupby('month')['delta'].sum()
month_delta_count = set_A.groupby('month')['delta'].count()
global_avg_delta  = set_A['delta'].mean()

def loo_month_delta(row):
    m = row['month']
    total = month_delta_sum.get(m, 0)
    count = month_delta_count.get(m, 0)
    if count <= 1:
        return global_avg_delta
    return (total - row['delta']) / (count - 1)

set_A['month_avg_delta'] = set_A.apply(loo_month_delta, axis=1)

# Year target encoding: leave-one-out mean delta by year
year_delta_sum   = set_A.groupby('year')['delta'].sum()
year_delta_count = set_A.groupby('year')['delta'].count()
```

```python
def loo_year_delta(row):
    y = row['year']
    total = year_delta_sum.get(y, 0)
    count = year_delta_count.get(y, 0)
    if count <= 1:
        return global_avg_delta
    return (total - row['delta']) / (count - 1)

set_A['year_avg_delta'] = set_A.apply(loo_year_delta, axis=1)

# Full Set A means
month_avg_delta_map = set_A.groupby('month')['delta'].mean()
year_avg_delta_map  = set_A.groupby('year')['delta'].mean()

def add_features(df, zipcode_stats, zipcode_delta_stats, global_avg_score,␣
 ↪global_avg_delta):
    df = df.merge(zipcode_stats, on='zipcode', how='left')
    df = df.merge(zipcode_delta_stats, on='zipcode', how='left')
    df['zipcode_avg_score']    = df['zipcode_avg_score'].
 ↪fillna(global_avg_score)
    df['zipcode_score_bin']    = (df['zipcode_avg_score'] // 10).astype(int)
    df['zipcode_score_std']    = df['zipcode_score_std'].fillna(global_std)
    df['zipcode_avg_delta']    = df['zipcode_avg_delta'].
 ↪fillna(global_avg_delta)
    df['score_deviation']      = df['prev_score'] - df['zipcode_avg_score']
    df['hist_score_mean']      = df['hist_score_mean'].fillna(df['prev_score'])
    df['hist_score_diff_mean'] = df['hist_score_diff_mean'].
 ↪fillna(global_avg_delta)
    df['hist_delta_std']       = df['hist_delta_std'].fillna(0.0)
    # month/year target encoding: use full Set A means (unseen months → global␣
 ↪mean)
    df['month_avg_delta'] = df['month'].map(month_avg_delta_map).
 ↪fillna(global_avg_delta)
    df['year_avg_delta']  = df['year'].map(year_avg_delta_map).
 ↪fillna(global_avg_delta)
    return df

set_B      = add_features(set_B,      zipcode_stats, zipcode_delta_stats,␣
 ↪global_avg, global_avg_delta)
pairs_test = add_features(pairs_test, zipcode_stats, zipcode_delta_stats,␣
 ↪global_avg, global_avg_delta)

print('Feature columns:', FEATURE_COLS)
print(f'Total features: {len(FEATURE_COLS)}')
print(set_A[FEATURE_COLS].isna().sum())
```

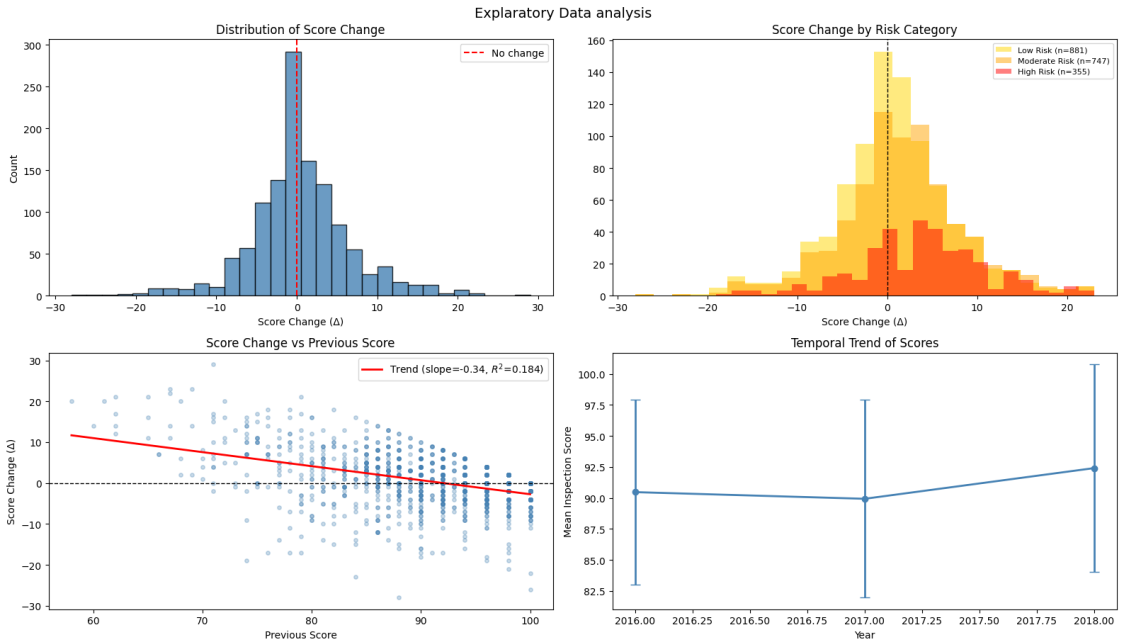
```
Feature columns: ['prev_score', 'has_low', 'has_moderate', 'has_high',
'num_violations', 'time_gap', 'zipcode_score_bin', 'zipcode_avg_delta',
'hist_score_diff_mean', 'hist_delta_std', 'month_avg_delta', 'year_avg_delta']
Total features: 12
prev_score              0
has_low                 0
has_moderate            0
has_high                0
num_violations          0
time_gap                0
zipcode_score_bin       0
zipcode_avg_delta       0
hist_score_diff_mean    0
hist_delta_std          0
month_avg_delta         0
year_avg_delta          0
dtype: int64
```

1.4 Step 0.3: Exploratory Data Analysis

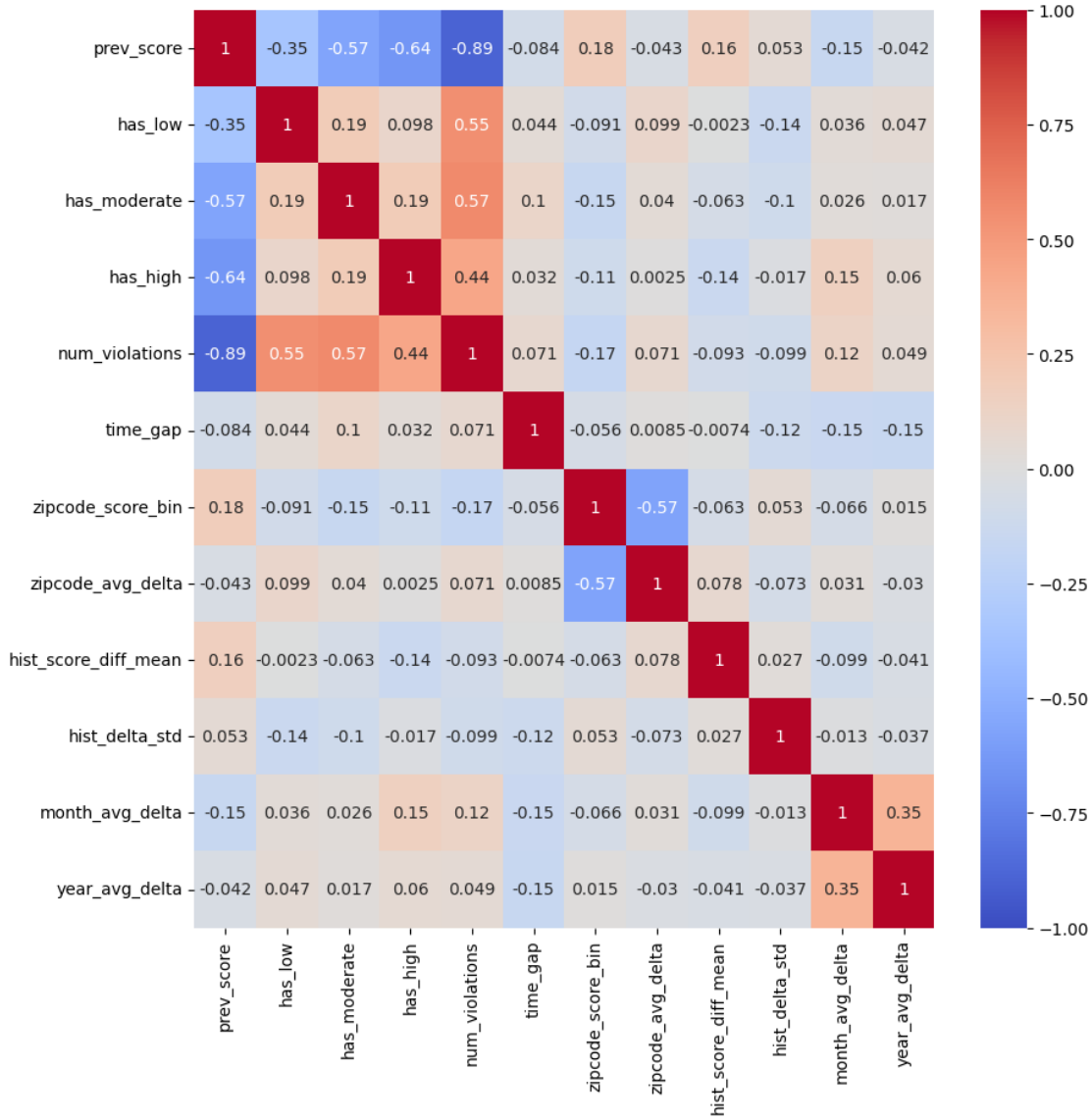
We use Set A for all EDA to avoid peeking at Set B.

[10]:
```
fig, axes = plt.subplots(2, 2, figsize=(16, 9))

# 1. Distribution of delta overall
axes[0, 0].hist(set_A['delta'], bins=30, edgecolor='black', color='steelblue',␣
↪alpha=0.8)
axes[0, 0].axvline(0, color='red', linestyle='--', linewidth=1.5, label='No␣
↪change')
axes[0, 0].set_xlabel('Score Change (Δ)')
axes[0, 0].set_ylabel('Count')
axes[0, 0].set_title('Distribution of Score Change')
axes[0, 0].legend()

# 2. Delta by risk category (motivates CQR: heteroscedasticity)
risk_labels = [
    ('has_low', 'Low Risk', 'gold'),
    ('has_moderate', 'Moderate Risk', 'orange'),
    ('has_high', 'High Risk', 'red')
]
for col, label, color in risk_labels:
    subset = set_A[set_A[col] == 1]['delta']
    axes[0, 1].hist(subset, bins=25, alpha=0.5, label=f'{label}␣
↪(n={len(subset)})', color=color)
axes[0, 1].set_xlabel('Score Change (Δ)')
axes[0, 1].set_title('Score Change by Risk Category')
```

```
axes[0, 1].legend(fontsize=8)
axes[0, 1].axvline(0, color='black', linestyle='--', linewidth=1)

# 3. Score change vs previous score (regression to mean)
axes[1, 0].scatter(set_A['prev_score'], set_A['delta'], alpha=0.3, s=15,␣
 ↪color='steelblue')
z = np.polyfit(set_A['prev_score'], set_A['delta'], 1)
p = np.poly1d(z)
xline = np.linspace(set_A['prev_score'].min(), set_A['prev_score'].max(), 100)
delta_pred = np.polyval(z, set_A['prev_score'])
ss_res = np.sum((set_A['delta'] - delta_pred) ** 2)
ss_tot = np.sum((set_A['delta'] - set_A['delta'].mean()) ** 2)
r2 = 1 - ss_res / ss_tot
axes[1, 0].plot(xline, p(xline), 'r-', linewidth=2, label=f'Trend (slope={z[0]:.
 ↪2f}, $R^2$={r2:.3f})')
axes[1, 0].axhline(0, color='black', linestyle='--', linewidth=1)
axes[1, 0].set_xlabel('Previous Score')
axes[1, 0].set_ylabel('Score Change (Δ)')
axes[1, 0].set_title('Score Change vs Previous Score')
axes[1, 0].legend()

# 4. Temporal trend: average score by year
yearly = set_A.groupby('year')['prev_score'].agg(['mean', 'std']).reset_index()
axes[1, 1].errorbar(yearly['year'], yearly['mean'], yerr=yearly['std'],
                    fmt='o-', capsize=5, color='steelblue', linewidth=2)
axes[1, 1].set_xlabel('Year')
axes[1, 1].set_ylabel('Mean Inspection Score')
axes[1, 1].set_title('Temporal Trend of Scores')

plt.tight_layout()
plt.suptitle('Explaratory Data analysis', fontsize=14, y=1.01)
plt.savefig('eda_plots.png', dpi=120, bbox_inches='tight')
plt.show()
```



[11]:
```
import seaborn as sns
plt.figure(figsize=(10, 10))
sns.heatmap(set_A[FEATURE_COLS].corr(), annot=True, cmap='coolwarm', vmin=-1,␣
↪vmax=1, center=0)
plt.show()
```



---

2 Part 1: Selective Inference

2.1 Step 1.1: Knockoffs — FDR-Controlled Feature Selection

We construct Gaussian Model-X knockoffs (Chp 2.6) on Set A to identify which features predict Δ, controlling FDR ≤ 0.2.

Assumption: Features $X \sim \mathcal{N}(\mu, \Sigma)$. We acknowledge this is an approximation and we fit a Bernoulli distribution to the binary features.

[12]:
```python
from sklearn.linear_model import LogisticRegression

def construct_gaussian_knockoffs(X, seed=42):
    """
    Gaussian equicorrelated knockoffs.

    Sampling via the conditional:
        X_tilde | X ~ N(mu + (I - D @ Sigma_inv) @ (x - mu),
                        2*D - D @ Sigma_inv @ D)
    """
    rng = np.random.default_rng(seed)
    n, p = X.shape

    # Step 1: estimate mu and Sigma
    mu    = X.mean(axis=0)
    Sigma = np.cov(X.T)

    # Step 2: choose D = diag(s) via equicorrelated rule
    Sigma_corr = np.corrcoef(X.T)
    s_scalar   = min(2 * np.linalg.eigvalsh(Sigma_corr).min(), 1.0) * 0.99
    s          = s_scalar * np.diag(Sigma)
    D          = np.diag(s)

    # Step 3: build conditional distribution X_tilde | X
    Sigma_inv  = np.linalg.pinv(Sigma)

    # conditional mean
    mu_cond    = mu + (X - mu) @ (np.eye(p) - Sigma_inv @ D).T

    # conditional covariance
    Sigma_cond = 2 * D - D @ Sigma_inv @ D
    Sigma_cond = (Sigma_cond + Sigma_cond.T) / 2
    eig_min    = np.linalg.eigvalsh(Sigma_cond).min()
    if eig_min < 0:
        Sigma_cond += (-eig_min + 1e-8) * np.eye(p)

    # Step 4: sample X_tilde for each row
    L       = np.linalg.cholesky(Sigma_cond)
    X_tilde = mu_cond + rng.standard_normal((n, p)) @ L.T

    return X_tilde, s, Sigma


def construct_mixed_knockoffs(X_std, X_raw, feature_names, seed=50):
    """
    Model-X knockoffs for mixed continuous / binary features.
```

```
    Binary features use sequential conditional sampling (SCIP construction):
        X_tilde_j ~ Bernoulli(P(X_j=1 | X_tilde_{<j}, X_{>j}))
    where X_tilde_{<j} are knockoffs already generated for earlier binary
    features and X_{>j} are original values for later ones.  Conditioning
    on the knockoff versions (rather than originals) for already-processed
    features is what makes the joint (X, X_tilde) satisfy the pairwise
    exchangeability property required by model-X knockoffs; using original
    values throughout would break this guarantee for correlated discrete
    features.

    Continuous features: Gaussian equicorrelated knockoffs on X_std
    (joint construction, so cross-feature correlations are handled exactly).
    """
    rng   = np.random.default_rng(seed)
    n, p  = X_std.shape
    X_tilde = np.empty_like(X_std)

    # Classify features from raw values
    binary_idx, cont_idx = [], []
    for j in range(p):
        col = X_raw[:, j]
        if set(np.unique(col[~np.isnan(col)].astype(float))).issubset({0.0, 1.
↪0}):
            binary_idx.append(j)
        else:
            cont_idx.append(j)

    print(f"  Binary features : {[feature_names[j] for j in binary_idx]}")
    print(f"  Continuous      : {[feature_names[j] for j in cont_idx]}")

    col_mean = X_raw.mean(axis=0)
    col_std  = X_raw.std(axis=0).clip(min=1e-8)

    # Sequential binary knockoffs
    X_mixed = X_raw.copy().astype(float)

    for j in binary_idx:
        col_raw = X_raw[:, j].astype(int)
        # Conditioning set: knockoffs for features processed before j,
        # originals for features processed after j (and all continuous).
        X_cond = np.delete(X_mixed, j, axis=1)
        if len(np.unique(col_raw)) < 2:
            p_hat = np.full(n, col_raw.mean())
        else:
            lr = LogisticRegression(max_iter=500, C=1.0, random_state=seed)
            lr.fit(X_cond, col_raw)
            p_hat = lr.predict_proba(X_cond)[:, 1]
```

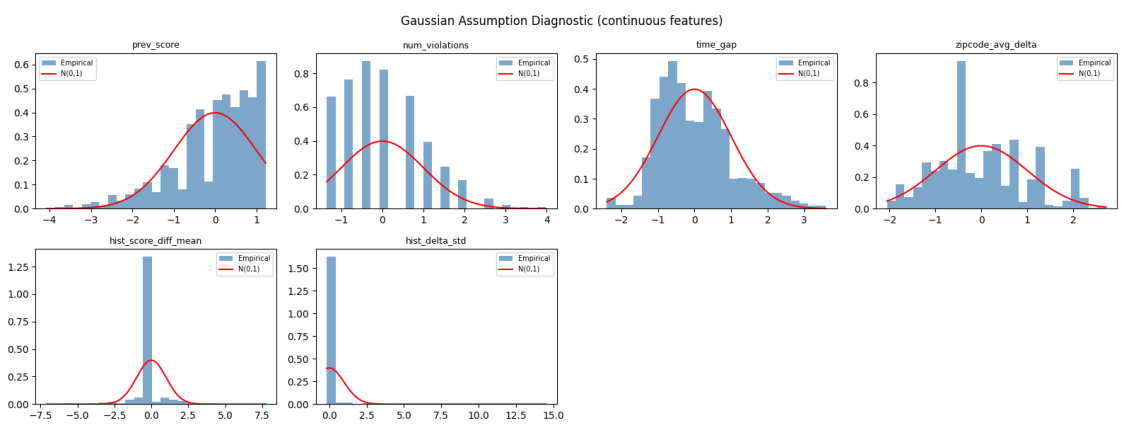
```
        knockoff_raw  = rng.binomial(1, p_hat).astype(float)
        X_tilde[:, j] = (knockoff_raw - col_mean[j]) / col_std[j]
        # Update X_mixed so the next binary feature conditions on this knockoff.
        X_mixed[:, j] = knockoff_raw

    # Continuous knockoffs: Gaussian equicorrelated on standardised block
    if cont_idx:
        Xt_cont, s_vals, Sigma = construct_gaussian_knockoffs(
            X_std[:, cont_idx], seed=seed)
        for k, j in enumerate(cont_idx):
            X_tilde[:, j] = Xt_cont[:, k]
    else:
        s_vals, Sigma = np.array([]), np.eye(1)

    return X_tilde, s_vals, Sigma, binary_idx, cont_idx


def knockoff_threshold_path(Z_orig, Z_tilde, fdr_level=0.2):
    """
    Knockoff threshold using Lasso path entry-time statistics.

    FDP_hat(t) = #{knockoffs in S_t} / #{originals in S_t}
    T = min{ t : FDP_hat(t) <= fdr_level }
    """
    candidates = np.sort(np.unique(np.concatenate([
        Z_orig[Z_orig > 0], Z_tilde[Z_tilde > 0]
    ])))
    for t in candidates:
        n_orig  = int(np.sum((Z_orig  >= t) & (Z_orig  > Z_tilde)))
        n_knock = int(np.sum((Z_tilde >= t) & (Z_tilde > Z_orig)))
        if n_orig == 0:
            continue
        if n_knock / n_orig <= fdr_level:
            return t, np.where((Z_orig >= t) & (Z_orig > Z_tilde))[0]
    return np.inf, np.array([], dtype=int)
```

[13]:
```
# Prepare Set A feature matrix
X_A = set_A[FEATURE_COLS].values.astype(float)
y_A = set_A['delta'].values.astype(float)

# Remove any rows with NaN
valid_mask = ~(np.isnan(X_A).any(axis=1) | np.isnan(y_A))
X_A, y_A   = X_A[valid_mask], y_A[valid_mask]
print(f'Set A after dropping NaN: {X_A.shape}')
1
# Standardize features
scaler = StandardScaler()
```

```
X_A_std = scaler.fit_transform(X_A)
```

```
Set A after dropping NaN: (1253, 12)
```

[14]:
```
# Diagnostic: validate Gaussian approximation for P_X
binary_feats = {'has_low', 'has_moderate', 'has_high', 'zipcode_score_bin',␣
 ↪'month_avg_delta', 'year_avg_delta'}
cont_cols = [f for f in FEATURE_COLS if f not in binary_feats]
cont_idx_diag = [FEATURE_COLS.index(f) for f in cont_cols]

ncols = 4
nrows = -(-len(cont_cols) // ncols)
fig, axes = plt.subplots(nrows, ncols, figsize=(ncols * 4, nrows * 3))
axes = axes.flatten()

for k, (feat, idx) in enumerate(zip(cont_cols, cont_idx_diag)):
    axes[k].hist(X_A_std[:, idx], bins=25, density=True,
                 color='steelblue', alpha=0.7, label='Empirical')
    xs = np.linspace(X_A_std[:, idx].min(), X_A_std[:, idx].max(), 200)
    axes[k].plot(xs, stats.norm.pdf(xs), 'r-', linewidth=1.5, label='N(0,1)')
    axes[k].set_title(feat, fontsize=9)
    axes[k].legend(fontsize=7)

for j in range(len(cont_cols), len(axes)):
    axes[j].set_visible(False)

plt.suptitle('Gaussian Assumption Diagnostic (continuous features)',␣
 ↪fontsize=12)
plt.tight_layout()
plt.savefig('knockoff_gaussian_diag.png', dpi=100, bbox_inches='tight')
plt.show()
```



[15]:
```
# Construct mixed knockoffs
print("Knockoff construction (mixed types):")
X_A_tilde, s_vals, Sigma_A, binary_idx, cont_idx = construct_mixed_knockoffs(
    X_A_std, X_A, FEATURE_COLS)
```

```
Knockoff construction (mixed types):
  Binary features : ['has_low', 'has_moderate', 'has_high']
  Continuous      : ['prev_score', 'num_violations', 'time_gap',
'zipcode_score_bin', 'zipcode_avg_delta', 'hist_score_diff_mean',
'hist_delta_std', 'month_avg_delta', 'year_avg_delta']
```

[16]:
```
from sklearn.linear_model import lasso_path

# Augmented feature matrix
X_aug = np.hstack([X_A_std, X_A_tilde])
p     = len(FEATURE_COLS)

# Compute full Lasso regularization path on the augmented matrix.
alphas, coefs_path, _ = lasso_path(X_aug, y_A, max_iter=10000, eps=1e-4)

Z = np.zeros(2 * p)
for j in range(2 * p):
    first_nonzero = np.where(np.abs(coefs_path[j, :]) > 1e-10)[0]
    if len(first_nonzero) > 0:
        Z[j] = alphas[first_nonzero[0]]  # highest lambda where coef is nonzero␣
 ↪= entry point

Z_orig  = Z[:p]   # entry lambdas for original features
Z_tilde = Z[p:]   # entry lambdas for knockoff copies
W_path  = Z_orig - Z_tilde  # positive = original entered before knockoff

print(f'Lasso path: {len(alphas)} steps, lambda range [{alphas[-1]:.5f},␣
 ↪{alphas[0]:.4f}]')
print(f'Features that never entered the model: {[FEATURE_COLS[j] for j in␣
 ↪range(p) if Z_orig[j] == 0]}')
print()
print(f'  {"Feature":25s}  {"Z_orig":8s}  {"Z_tilde":8s}  {"W = Z-Z":8s} ␣
 ↪{"entered first"}')
print('  ' + '-'*72)
for feat, zo, zt, w in zip(FEATURE_COLS, Z_orig, Z_tilde, W_path):
    winner = 'ORIGINAL' if zo > zt else ('knockoff' if zt > zo else 'tie')
    print(f'  {feat:25s}  {zo:8.5f}  {zt:8.5f}  {w:+8.5f}  {winner}')
```

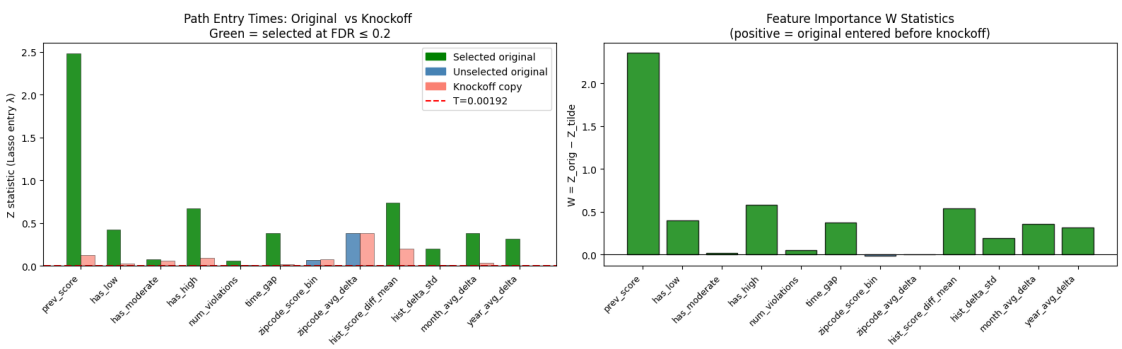
```
Lasso path: 100 steps, lambda range [0.00027, 2.7226]
Features that never entered the model: []

  Feature                    Z_orig    Z_tilde   W = Z-Z  entered first
  ------------------------------------------------------------------------
```

```
 prev_score                  2.48077   0.12637  +2.35440  ORIGINAL
 has_low                     0.42356   0.02599  +0.39757  ORIGINAL
 has_moderate                0.07937   0.06004  +0.01933  ORIGINAL
 has_high                    0.67442   0.09560  +0.57882  ORIGINAL
 num_violations              0.06004   0.00934  +0.05070  ORIGINAL
 time_gap                    0.38593   0.01487  +0.37106  ORIGINAL
 zipcode_score_bin           0.06589   0.07937  -0.01348  knockoff
 zipcode_avg_delta           0.38593   0.38593  +0.00000  tie
 hist_score_diff_mean        0.74018   0.20122  +0.53895  ORIGINAL
 hist_delta_std              0.20122   0.00934  +0.19188  ORIGINAL
 month_avg_delta             0.38593   0.03130  +0.35462  ORIGINAL
 year_avg_delta              0.32040   0.00192  +0.31848  ORIGINAL
```

[17]:
```
from matplotlib.patches import Patch

# Apply knockoff threshold
T_knockoff, selected_idx = knockoff_threshold_path(Z_orig, Z_tilde,␣
 ↪fdr_level=FDR_LEVEL)
selected_features = [FEATURE_COLS[i] for i in selected_idx]

print(f'Knockoff threshold T = {T_knockoff}')
print(f'Selected features (FDR <= {FDR_LEVEL}): {selected_features}')
print(f'Number of selected features: {len(selected_features)}')

p   = len(FEATURE_COLS)
x   = np.arange(p)
w   = 0.35
fig, axes = plt.subplots(1, 2, figsize=(16, 5))

for idx in range(p):
    c_o = 'green' if idx in selected_idx else 'steelblue'
    axes[0].bar(idx - w/2, Z_orig[idx],  width=w, color=c_o,     alpha=0.85,␣
 ↪edgecolor='black', linewidth=0.4)
    axes[0].bar(idx + w/2, Z_tilde[idx], width=w, color='salmon', alpha=0.70,␣
 ↪edgecolor='black', linewidth=0.4)
if np.isfinite(T_knockoff):
    axes[0].axhline(T_knockoff, color='red', linestyle='--', linewidth=2,
                    label=f'Threshold T = {T_knockoff:.5f}')
    axes[0].legend()
axes[0].set_xticks(x)
axes[0].set_xticklabels(FEATURE_COLS, rotation=45, ha='right', fontsize=9)
axes[0].set_ylabel('Z statistic (Lasso entry  )')
axes[0].set_title(f'Path Entry Times: Original  vs Knockoff \nGreen = selected␣
 ↪at FDR  {FDR_LEVEL}')
axes[0].legend(handles=[
    Patch(color='green',   label='Selected original'),
    Patch(color='steelblue', label='Unselected original'),
```

```
    Patch(color='salmon',  label='Knockoff copy'),
] + ([plt.Line2D([0],[0], color='red', linestyle='--', label=f'T={T_knockoff:.
 ↪5f}')] if np.isfinite(T_knockoff) else []))

# Right: W = Z_orig - Z_tilde
colors_w = ['green' if i in selected_idx else 'steelblue' for i in range(p)]
axes[1].bar(x, W_path, color=colors_w, edgecolor='black', alpha=0.8)
axes[1].axhline(0, color='black', linewidth=0.8)
axes[1].set_xticks(x)
axes[1].set_xticklabels(FEATURE_COLS, rotation=45, ha='right', fontsize=9)
axes[1].set_ylabel('W = Z_orig - Z_tilde')
axes[1].set_title('Feature Importance W Statistics\n(positive = original␣
 ↪entered before knockoff)')

plt.tight_layout()
plt.savefig('knockoff_W_stats.png', dpi=120, bbox_inches='tight')
plt.show()
```

```
Knockoff threshold T = 0.001920773633277867
Selected features (FDR <= 0.2): ['prev_score', 'has_low', 'has_moderate',
'has_high', 'num_violations', 'time_gap', 'hist_score_diff_mean',
'hist_delta_std', 'month_avg_delta', 'year_avg_delta']
Number of selected features: 10
```



[18]:
```
# --- Pairwise correlation: each original Xj vs its knockoff Xj ---
pair_corr = pd.Series(
    [np.corrcoef(X_A_std[:, j], X_A_tilde[:, j])[0, 1] for j in␣
 ↪range(len(FEATURE_COLS))],
    index=FEATURE_COLS,
    name='corr(Xj, Xj)'
)

print(pair_corr.round(3).to_string())
```

```
prev_score              0.800
```

```
has_low                 0.715
has_moderate            0.508
has_high                0.637
num_violations          0.808
time_gap                0.801
zipcode_score_bin       0.796
zipcode_avg_delta       0.795
hist_score_diff_mean    0.810
hist_delta_std          0.794
month_avg_delta         0.803
year_avg_delta          0.804
```

2.2 Step 1.2: FCR-Controlling Confidence Intervals

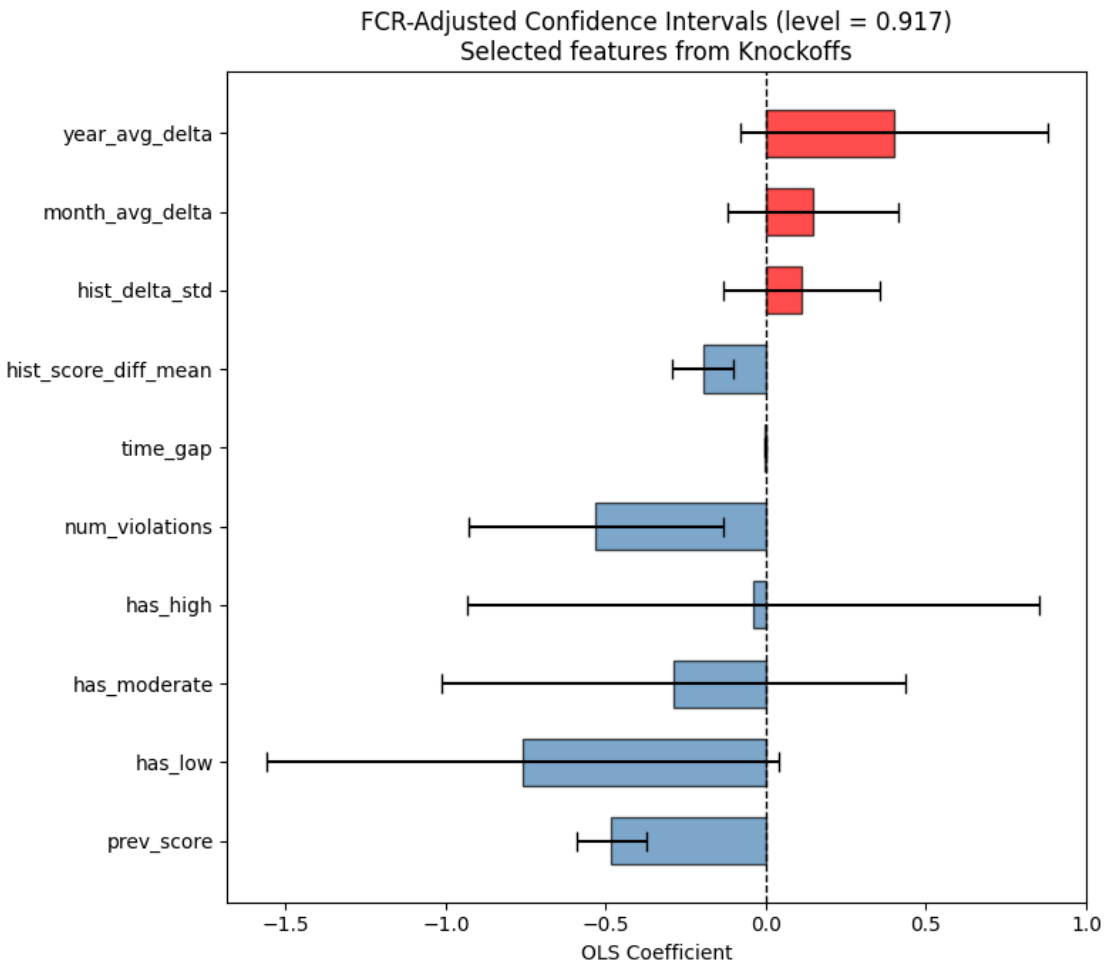
After knockoffs select $\hat{S}$, we report confidence intervals for the regression coefficients β_j, $j \in \hat{S}$, at the FCR-adjusted level $1 - \alpha\hat{K}/m$, where $\hat{K} = |\hat{S}|$ and m is the total number of features.

[19]:
```
m      = len(FEATURE_COLS)
K_hat  = len(selected_features)
alpha_fcr = FCR_ALPHA * K_hat / m

print(f'FCR adjustment: alpha_fcr = {FCR_ALPHA} * {K_hat} / {m} = {alpha_fcr:.
 ↪4f}')
print(f'CI level: 1 - alpha_fcr = {1 - alpha_fcr:.4f}\n')

if K_hat > 0:
    X_sel = X_A[:, selected_idx]   # use unstandardized for interpretability
    X_sel_const = sm.add_constant(X_sel)
    ols_model = sm.OLS(y_A, X_sel_const).fit()

    ci = ols_model.conf_int(alpha=alpha_fcr)
    coef_names = ['intercept'] + selected_features

    print(f'OLS on selected features with FCR-adjusted CIs (level =␣
 ↪{1-alpha_fcr:.3f}):')
    print(f'{"Feature":25s}  {"Coef":8s}  {"CI Lower":10s}  {"CI Upper":10s} ␣
 ↪{"p-value":8s}')
    print('-' * 70)
    for i, name in enumerate(coef_names):
        print(f'{name:25s}  {ols_model.params[i]:8.4f}  '
              f'{ci[i, 0]:10.4f}  {ci[i, 1]:10.4f}  '
              f'{ols_model.pvalues[i]:8.4f}')
    print(f'\nModel R²: {ols_model.rsquared:.4f}')
```

```
FCR adjustment: alpha_fcr = 0.1 * 10 / 12 = 0.0833
CI level: 1 - alpha_fcr = 0.9167
```

```
OLS on selected features with FCR-adjusted CIs (level = 0.917):
Feature                     Coef      CI Lower    CI Upper    p-value
-----------------------------------------------------------------------
intercept                  46.8647     35.9061     57.8234    0.0000
prev_score                 -0.4818     -0.5913     -0.3724    0.0000
has_low                    -0.7593     -1.5602      0.0415    0.1006
has_moderate               -0.2880     -1.0106      0.4346    0.4898
has_high                   -0.0394     -0.9333      0.8544    0.9391
num_violations             -0.5313     -0.9285     -0.1342    0.0206
time_gap                   -0.0030     -0.0052     -0.0007    0.0242
hist_score_diff_mean       -0.1958     -0.2909     -0.1008    0.0004
hist_delta_std              0.1118     -0.1311      0.3547    0.4252
month_avg_delta             0.1461     -0.1197      0.4118    0.3410
year_avg_delta              0.4004     -0.0782      0.8790    0.1474

Model R²: 0.2183
```

[20]:
```python
# Visualize FCR CIs
if K_hat > 0:
    fig, ax = plt.subplots(figsize=(8, max(4, K_hat * 0.5 + 2)))
    coefs_plot  = ols_model.params[1:]   # exclude intercept
    ci_plot     = ci[1:]             # exclude intercept
    y_pos = range(len(selected_features))

    ax.barh(y_pos, coefs_plot,
            xerr=[coefs_plot - ci_plot[:, 0], ci_plot[:, 1] - coefs_plot],
            color=['red' if c > 0 else 'steelblue' for c in coefs_plot],
            alpha=0.7, capsize=5, height=0.6, edgecolor='black')
    ax.axvline(0, color='black', linewidth=1, linestyle='--')
    ax.set_yticks(y_pos)
    ax.set_yticklabels(selected_features)
    ax.set_xlabel('OLS Coefficient')
    ax.set_title(f'FCR-Adjusted Confidence Intervals (level = {1-alpha_fcr:.
 ↪3f})\nSelected features from Knockoffs')
    plt.tight_layout()
    plt.savefig('fcr_cis.png', dpi=120, bbox_inches='tight')
    plt.show()
```



---

3 Part 2: Predictive Inference

3.1 Step 2.1: Conformalized Prediction Intervals

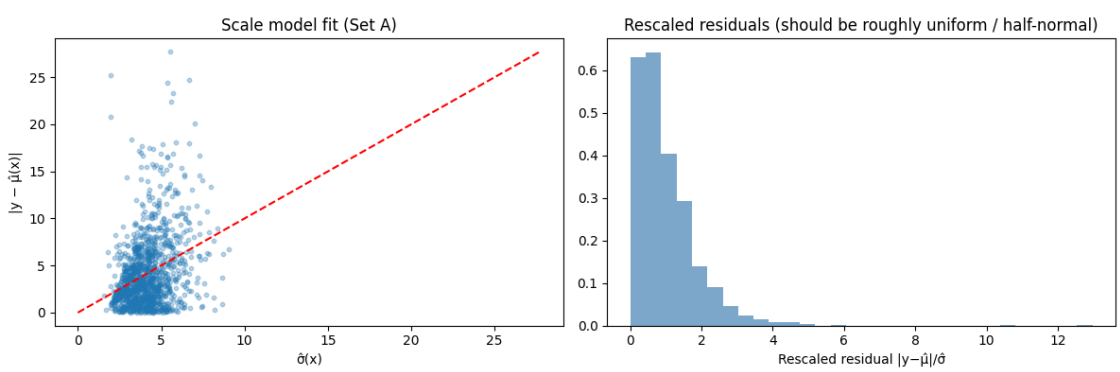
We reuse the OLS model $\hat{\mu}(x)$ from Phase 1 for prediction.

From **PSet5 §3b**: fit $\hat{\mu}$ and $\hat{}$ on the training data, calibrate rescaled residuals $|y - \hat{\mu}(x)| / \hat{}(x)$ on held-out data, and output adaptive confidence intervals.

[21]:
```
from sklearn.linear_model import LinearRegression

#   Prepare feature matrices (unstandardised, selected features only)
X_A_sel  = X_A[:, selected_idx]
X_B_raw  = set_B[FEATURE_COLS].values.astype(float)
y_B      = set_B['delta'].values.astype(float)
rest_ids_B = set_B['business_id'].values
```

```
valid_B  = ~(np.isnan(X_B_raw).any(axis=1) | np.isnan(y_B))
X_B_raw, y_B, rest_ids_B = X_B_raw[valid_B], y_B[valid_B], rest_ids_B[valid_B]
X_B_sel  = X_B_raw[:, selected_idx]

#   μ̂: predictions from the Phase-1 OLS model
mu_hat_A  = ols_model.predict(sm.add_constant(X_A_sel,  has_constant='add'))
mu_hat_B  = ols_model.predict(sm.add_constant(X_B_sel,  has_constant='add'))

#   ^: fit conditional scale on Set A residuals (same as PSet5 §3b)
abs_resid_A = np.abs(y_A - mu_hat_A)
sigma_model = LinearRegression().fit(X_A_sel, abs_resid_A)

sigma_A = np.maximum(sigma_model.predict(X_A_sel), 1e-3)
sigma_B = np.maximum(sigma_model.predict(X_B_sel), 1e-3)

print(f'Set B calibration size : {len(y_B)}')
print(f'sigma_hat (Set B): mean={sigma_B.mean():.3f}, std={sigma_B.std():.3f}')

# Diagnostic: sigma_hat vs |residual| on Set A
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
axes[0].scatter(sigma_A, abs_resid_A, alpha=0.3, s=10)
axes[0].plot([0, abs_resid_A.max()], [0, abs_resid_A.max()], 'r--')
axes[0].set_xlabel('^(x)'); axes[0].set_ylabel('|y - μ̂(x)|')
axes[0].set_title('Scale model fit (Set A)')

rescaled_A = abs_resid_A / sigma_A
axes[1].hist(rescaled_A, bins=30, density=True, color='steelblue', alpha=0.7)
axes[1].set_xlabel('Rescaled residual |y-μ̂|/^')
axes[1].set_title('Rescaled residuals (should be roughly uniform /␣
 ↪half-normal)')
plt.tight_layout(); plt.show()
```

```
Set B calibration size : 1266
sigma_hat (Set B): mean=4.099, std=1.194
```



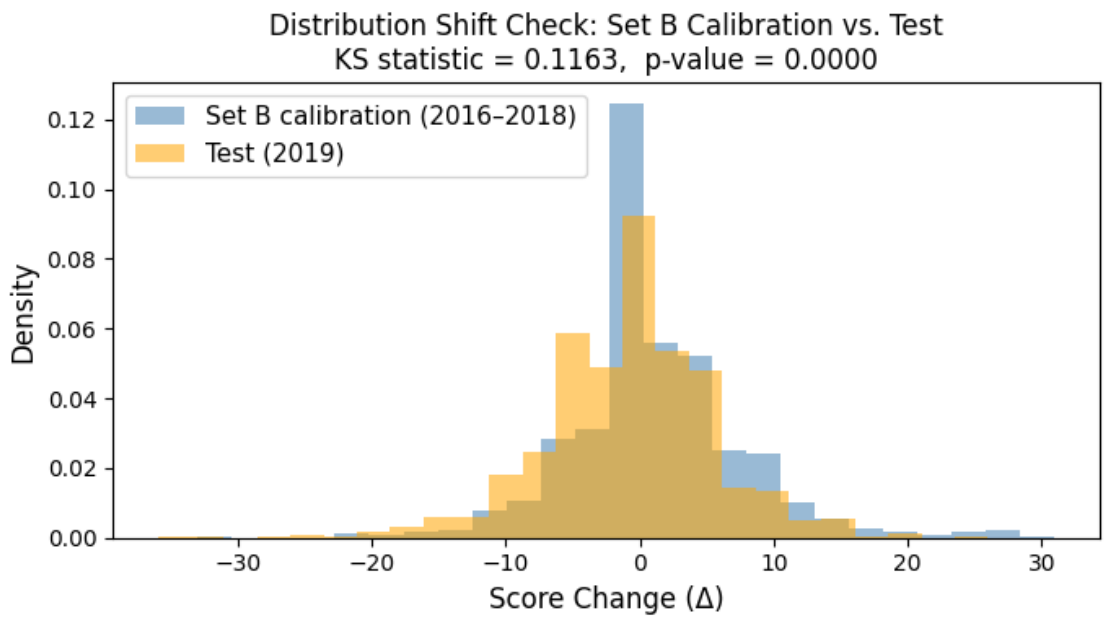
[22]:
```
#   Calibrate rescaled residuals on Set B
rescaled_B = np.abs(y_B - mu_hat_B) / sigma_B
n_cal      = len(rescaled_B)
q_hat      = np.quantile(rescaled_B,
                         (1 - COVERAGE_ALPHA) * (1 + 1 / n_cal),
                         method='higher')

print(f'Calibration quantile q_hat = {q_hat:.4f}')
print(f'Rescaled residuals: mean={rescaled_B.mean():.3f}, '
      f'90th pct={np.quantile(rescaled_B, 0.9):.3f}')

#   Plain (non-rescaled) conformal as baseline
plain_resid_B = np.abs(y_B - mu_hat_B)
q_hat_plain   = np.quantile(plain_resid_B,
                            (1 - COVERAGE_ALPHA) * (1 + 1 / n_cal),
                            method='higher')
print(f'Plain conformal q_hat      = {q_hat_plain:.4f}')
```

```
Calibration quantile q_hat = 2.3081
Rescaled residuals: mean=0.961, 90th pct=1.906
Plain conformal q_hat      = 10.5305
```

3.2 Step 2.2: Conformalized Quantile Regression (CQR)

[23]:
```
#   Distribution Shift Check (Set B calibration vs. 2019 test)
all_for_ks = set_B['delta'].dropna()
test_ks    = pairs_test['delta'].dropna()

ks_stat, ks_pval = stats.ks_2samp(all_for_ks, test_ks)

fig, ax = plt.subplots(figsize=(7, 4))
ax.hist(all_for_ks, bins=25, alpha=0.55, label='Set B calibration (2016-2018)',
        color='steelblue', density=True)
ax.hist(test_ks,    bins=25, alpha=0.55, label='Test (2019)',
        color='orange',    density=True)
ax.set_xlabel('Score Change (Δ)', fontsize=12)
ax.set_ylabel('Density', fontsize=12)
ax.set_title(
    f'Distribution Shift Check: Set B Calibration vs. Test\n'
    f'KS statistic = {ks_stat:.4f},  p-value = {ks_pval:.4f}',
    fontsize=12
)
ax.legend(fontsize=11)
plt.tight_layout()
```

```
plt.savefig('distribution_shift_check.png', dpi=120, bbox_inches='tight')
plt.show()

print(f'\nKS test (Set B vs 2019): statistic={ks_stat:.4f}, p-value={ks_pval:.
 ↪4f}')
if ks_pval < 0.05:
    print('→ Evidence of distribution shift between calibration and test␣
 ↪samples.')
    print('  Exact exchangeability is questionable for 2019 coverage evaluation.
 ↪')
    print('  Adaptive methods (CQR, kNN conformal) may improve empirical␣
 ↪performance.')
else:
    print('→ No strong evidence of shift. Exchangeability looks more plausible.
 ↪')

#   Test set setup
X_test_all = pairs_test[FEATURE_COLS].values.astype(float)
y_test_raw = pairs_test['delta'].values.astype(float)
valid_test  = ~(np.isnan(X_test_all).any(axis=1) | np.isnan(y_test_raw))
X_test_all, y_test = X_test_all[valid_test], y_test_raw[valid_test]
pairs_test_valid = pairs_test[valid_test].copy()
X_test_sel = X_test_all[:, selected_idx]

mu_hat_te  = ols_model.predict(sm.add_constant(X_test_sel, has_constant='add'))
sigma_te   = np.maximum(sigma_model.predict(X_test_sel), 1e-3)

# Rescaled and plain intervals
lower_rs = mu_hat_te - q_hat       * sigma_te
upper_rs = mu_hat_te + q_hat       * sigma_te
lower_pl = mu_hat_te - q_hat_plain * np.ones(len(mu_hat_te))
upper_pl = mu_hat_te + q_hat_plain * np.ones(len(mu_hat_te))
```




```
KS test (Set B vs 2019): statistic=0.1163, p-value=0.0000
→ Evidence of distribution shift between calibration and test samples.
  Exact exchangeability is questionable for 2019 coverage evaluation.
  Adaptive methods (CQR, kNN conformal) may improve empirical performance.
```

[24]:
```
#  Conformalized Quantile Regression (CQR)

X_A_const  = sm.add_constant(X_A_sel,    has_constant='add')
X_B_const  = sm.add_constant(X_B_sel,    has_constant='add')
X_te_const = sm.add_constant(X_test_sel, has_constant='add')

q_lo_model = QuantReg(y_A, X_A_const).fit(q=COVERAGE_ALPHA / 2,
 ↪max_iter=3000)
q_hi_model = QuantReg(y_A, X_A_const).fit(q=1 - COVERAGE_ALPHA / 2,
 ↪max_iter=3000)

# Calibrate CQR score on Set B
gamma_lo_B   = q_lo_model.predict(X_B_const)
gamma_hi_B   = q_hi_model.predict(X_B_const)
cqr_scores_B = np.maximum(y_B - gamma_hi_B, gamma_lo_B - y_B)
q_hat_cqr    = np.quantile(cqr_scores_B,
                           (1 - COVERAGE_ALPHA) * (1 + 1 / n_cal),
                           method='higher')

# Test set intervals
```

```
gamma_lo_te = q_lo_model.predict(X_te_const)
gamma_hi_te = q_hi_model.predict(X_te_const)
lower_cqr   = gamma_lo_te - q_hat_cqr
upper_cqr   = gamma_hi_te + q_hat_cqr

print(f'CQR calibration quantile q̂  = {q_hat_cqr:.4f}')
print(f'CQR avg interval width       = {(upper_cqr - lower_cqr).mean():.4f}')
print(f'Quantile lo (test): mean={gamma_lo_te.mean():.3f}   '
      f'Quantile hi (test): mean={gamma_hi_te.mean():.3f}')
```

```
CQR calibration quantile q̂  = -0.0000
CQR avg interval width       = 22.9287
Quantile lo (test): mean=-13.571   Quantile hi (test): mean=9.357
```

3.3 Step 2.3: Local (k-Nearest-Neighbor) Conformal Prediction

Instead of a single global quantile over all calibration points, we compute a **local quantile** for each test point from its k nearest neighbors in the calibration set.

Procedure: 1. For each test point x_{te}, find its k nearest neighbors $\mathcal{N}(x_{\text{te}}) \subseteq \mathcal{B}$ (standardised feature space). 2. Compute the local conformal quantile from the k neighbors' rescaled nonconformity scores $s_i = |y_i - \hat{\mu}(x_i)|/\hat{\sigma}(x_i)$. 3. Prediction interval: $\hat{\mu}(x_{\text{te}}) \pm \hat{q}_k(x_{\text{te}}) \cdot \hat{\sigma}(x_{\text{te}})$.

[25]:
```
from sklearn.neighbors import NearestNeighbors
from sklearn.preprocessing import StandardScaler

# Standardise selected features for distance computation
scaler_knn  = StandardScaler().fit(X_B_sel)
X_B_scaled  = scaler_knn.transform(X_B_sel)
X_te_scaled = scaler_knn.transform(X_test_sel)

# Try different values of k
K_values = [20, 30, 50, 100, 200, 500, 1000]

knn_results = {}
print(f'kNN local conformal (target coverage >= {1-COVERAGE_ALPHA:.2f})')
print(f'{"k":>6}  {"coverage":>9}  {"avg width":>10}')
print('-' * 32)

for K in K_values:
    nn = NearestNeighbors(n_neighbors=K).fit(X_B_scaled)
    _, idx = nn.kneighbors(X_te_scaled)   # (n_test, K)

    level   = (1 - COVERAGE_ALPHA) * (1 + 1 / K)
    q_local = np.array([
        np.quantile(rescaled_B[idx[i]], level, method='higher')
        for i in range(len(X_te_scaled))
```

```
    ])

    lower_k = mu_hat_te - q_local * sigma_te
    upper_k = mu_hat_te + q_local * sigma_te

    covered = (y_test >= lower_k) & (y_test <= upper_k)
    cov   = covered.mean()
    width = (upper_k - lower_k).mean()
    print(f'{K:>6}  {cov:>9.4f}  {width:>10.4f}')
    knn_results[K] = dict(lower=lower_k, upper=upper_k,
                          covered=covered, cov=cov, width=width,
                          q_local=q_local)

# Pick k closest to nominal coverage, here we may have to consider some␣
 ↪multiplicity correction
K_best      = min(K_values, key=lambda k: abs(knn_results[k]['cov'] - (1 -␣
 ↪COVERAGE_ALPHA)))
lower_knn   = knn_results[K_best]['lower']
upper_knn   = knn_results[K_best]['upper']
cov_knn     = knn_results[K_best]['cov']
width_knn   = knn_results[K_best]['width']
covered_knn = knn_results[K_best]['covered']
q_local_best = knn_results[K_best]['q_local']
print(f'\nSelected k = {K_best}  (coverage = {cov_knn:.4f}, avg width =␣
 ↪{width_knn:.4f})')
```

```
kNN local conformal (target coverage >= 0.95)
     k   coverage   avg width
--------------------------------
    20     0.9318     25.7053
    30     0.9527     28.6224
    50     0.9384     23.7188
   100     0.9208     20.0272
   200     0.9247     19.7201
   500     0.9263     19.6814
  1000     0.9258     19.5361

Selected k = 30  (coverage = 0.9527, avg width = 28.6224)
```

3.4 Part 3: Evaluation on 2019 Test Set

[26]:
```
def coverage_and_width(lower, upper, y_true, label=""):
    covered = (y_true >= lower) & (y_true <= upper)
    cov   = covered.mean()
    width = (upper - lower).mean()
    print(f'{label:45s}: coverage = {cov:.4f},  avg width = {width:.4f}')
```

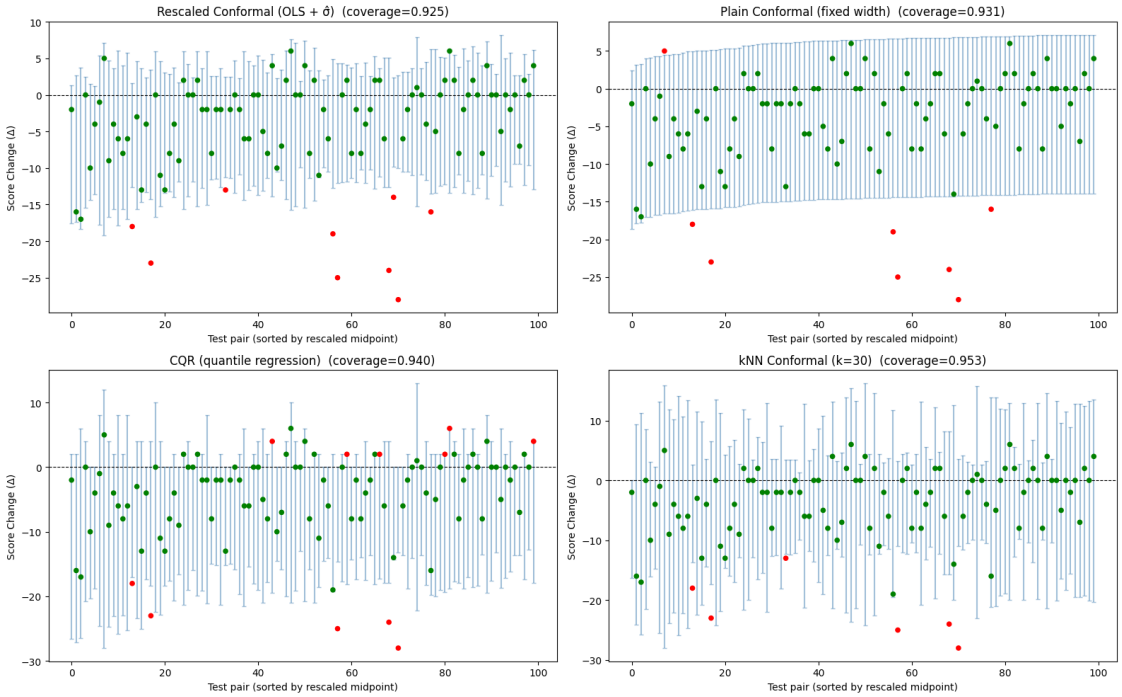
```
    return covered, cov, width

print("=" * 78)
print(f"Target coverage = {1-COVERAGE_ALPHA:.2f}   (exact under exchangeability␣
↪for split conformal)")
print("=" * 78)

covered_rs,  cov_rs,  width_rs  = coverage_and_width(lower_rs,  upper_rs, ␣
↪y_test,
                                                      "Rescaled conformal (OLS␣
↪+ ^)")
covered_pl,  cov_pl,  width_pl  = coverage_and_width(lower_pl,  upper_pl, ␣
↪y_test,
                                                      "Plain conformal (fixed␣
↪width)")
covered_cqr, cov_cqr, width_cqr = coverage_and_width(lower_cqr, upper_cqr,␣
↪y_test,
                                                      "CQR (conformalized␣
↪quantile regression)")
covered_knn2, cov_knn2, width_knn2 = coverage_and_width(lower_knn, upper_knn,␣
↪y_test,
                                                         f"kNN conformal␣
↪(k={K_best})")
```

```
==============================================================================
Target coverage = 0.95   (exact under exchangeability for split conformal)
==============================================================================
Rescaled conformal (OLS + ^)                : coverage = 0.9252,  avg width =
19.4110
Plain conformal (fixed width)                : coverage = 0.9313,  avg width =
21.0610
CQR (conformalized quantile regression)      : coverage = 0.9395,  avg width =
22.9287
kNN conformal (k=30)                         : coverage = 0.9527,  avg width =
28.6224
```

[27]:
```
# Coverage by risk category
print('\nCoverage by risk category (Rescaled conformal):')
for col, name in zip(['has_low','has_moderate','has_high'],
                     ['Low Risk','Moderate Risk','High Risk']):
    mask = pairs_test_valid[col].values.astype(bool)
    if mask.sum() == 0: continue
    cov_r = covered_rs[mask].mean()
    w_r   = (upper_rs[mask] - lower_rs[mask]).mean()
    print(f'  {name:15s}: coverage={cov_r:.4f}, avg width={w_r:.4f}, n={mask.
↪sum()}')
```

```
Coverage by risk category (Rescaled conformal):
  Low Risk       : coverage=0.9318, avg width=21.1671, n=1290
  Moderate Risk  : coverage=0.9298, avg width=21.4879, n=1139
  High Risk      : coverage=0.9150, avg width=23.0893, n=506
```

[28]:
```
# Visualisation: prediction intervals vs actual delta - all four methods
methods = [
    ('Rescaled Conformal (OLS + ^)', lower_rs,  upper_rs,  covered_rs,  cov_rs),
    ('Plain Conformal (fixed width)',  lower_pl,  upper_pl,  covered_pl, ␣
 ↪cov_pl),
    ('CQR (quantile regression)',      lower_cqr, upper_cqr, covered_cqr,␣
 ↪cov_cqr),
    (f'kNN Conformal (k={K_best})',    lower_knn, upper_knn, covered_knn,␣
 ↪cov_knn2),
]

# Common ordering across all panels (sort by rescaled midpoint)
mid_ref = (lower_rs + upper_rs) / 2
order   = np.argsort(mid_ref)[:100]

fig, axes = plt.subplots(2, 2, figsize=(16, 10))

for ax, (title, lo, hi, covered, cov) in zip(axes.flat, methods):
    mid = (lo + hi) / 2
    ax.errorbar(range(len(order)), mid[order],
                yerr=[mid[order]-lo[order], hi[order]-mid[order]],
                fmt="none", ecolor="steelblue", alpha=0.5, capsize=2)
    ax.scatter(range(len(order)), y_test[order],
               c=["green" if c else "red" for c in covered[order]],
               s=20, zorder=5)
    ax.axhline(0, color="black", linewidth=0.8, linestyle="--")
    ax.set_xlabel("Test pair (sorted by rescaled midpoint)")
    ax.set_ylabel("Score Change (Δ)")
    ax.set_title(f"{title}  (coverage={cov:.3f})")

plt.tight_layout()
plt.savefig("evaluation_results.png", dpi=120, bbox_inches="tight")
plt.show()
```



[30]:
```
print("=" * 68)
print("FINAL RESULTS SUMMARY")
print("=" * 68)
print(f"\Part 1 - Knockoffs (FDR  {FDR_LEVEL})")
print(f"  Selected features : {selected_features}")
print(f"  Threshold T       = {T_knockoff:.4f}")
print(f"\nPart 1 - FCR CIs (level {1-alpha_fcr:.3f})")
if K_hat > 0:
    print(f"  OLS R²            = {ols_model.rsquared:.4f}")

print(f"\nPhase 2 - Prediction Intervals on 2019 Test Set")
print(f"  {'Method':<45} {'Coverage':>9} {'Avg Width':>11}")
print(f"  {'-'*67}")
for label, cov, width in [
    ("Rescaled conformal (OLS + ^)",              cov_rs,  width_rs),
    ("Plain conformal (fixed width)",              cov_pl,  width_pl),
    ("CQR (conformalized quantile regression)",    cov_cqr, width_cqr),
    (f"kNN conformal (k={K_best})",                cov_knn, width_knn)
]:
    print(f"  {label:<45} {cov:>9.4f} {width:>11.4f}")
```

```
====================================================================
FINAL RESULTS SUMMARY
====================================================================
\Part 1 - Knockoffs (FDR  0.2)
```

```
  Selected features : ['prev_score', 'has_low', 'has_moderate', 'has_high',
'num_violations', 'time_gap', 'hist_score_diff_mean', 'hist_delta_std',
'month_avg_delta', 'year_avg_delta']
  Threshold T       = 0.0019

Part 1 - FCR CIs (level 0.917)
  OLS R²            = 0.2183

Phase 2 - Prediction Intervals on 2019 Test Set
  Method                                          Coverage   Avg Width
  ---------------------------------------------------------------------
  Rescaled conformal (OLS + ^)                     0.9252     19.4110
  Plain conformal (fixed width)                     0.9313     21.0610
  CQR (conformalized quantile regression)           0.9395     22.9287
  kNN conformal (k=30)                              0.9527     28.6224
```